

SPR NEAR GAUSSIAN

1. AN ALTERNATIVE APPROACH TO STABLE STFT PHASE RETRIEVAL

In this note, we present an alternative route which proves a weaker version of the local stable phase retrieval theorem near the Gaussian. This argument bypasses – at the cost of restricting ourselves to the class of functions which are *squares* – the circle estimates and the frequency localization used in the main approach, replacing them with a direct integration by parts and a weighted Cauchy integral inequality.

Throughout this note, $F, G \in \mathcal{F}^2(\mathbb{C})$ denote polynomials unless stated otherwise; the conclusion then extends to arbitrary squares in $\mathcal{F}^2(\mathbb{C})$ by a standard density argument. We abbreviate

$$\mathcal{F}_*^2(\mathbb{C}) := \{H \in \mathcal{F}^2(\mathbb{C}) : H(0) = 0\},$$

which is the closed subspace of $\mathcal{F}^2(\mathbb{C})$ consisting of functions orthogonal to the constant function.

1.0.1. An equivalent norm. The starting point is a weighted Hardy-type identity that expresses the Fock norm of an element of $\mathcal{F}_*^2(\mathbb{C})$ in terms of a weighted L^2 norm of its derivative.

Lemma 1.1. *There exist absolute constants $c_1, c_2 \in (0, \infty)$ such that for every $H \in \mathcal{F}_*^2(\mathbb{C})$,*

$$c_1 \|H\|_{\mathcal{F}}^2 \leq \int_{\mathbb{C}} |H'(z)|^2 \frac{d\gamma(z)}{1 + |z|^2} \leq c_2 \|H\|_{\mathcal{F}}^2. \quad (1.1)$$

Proof. We expand $H(z) = \sum_{n \geq 1} a_n z^n$ with $\sum_{n \geq 1} |a_n|^2 n! = \|H\|_{\mathcal{F}}^2$. Since the monomials z^n are pairwise orthogonal in $L^2(d\gamma)$, the integral in (1.1) decomposes as

$$\int_{\mathbb{C}} |H'(z)|^2 \frac{d\gamma(z)}{1 + |z|^2} = \sum_{n \geq 1} n^2 |a_n|^2 \kappa_n, \quad \kappa_n := \int_{\mathbb{C}} \frac{|z|^{2(n-1)}}{1 + |z|^2} d\gamma(z).$$

Writing $1/(1 + |z|^2) = \int_0^\infty e^{-(1+|z|^2)s} ds$ and using the identity $\int_{\mathbb{C}} |z|^{2k} d\gamma = k!$, we have

$$\kappa_n = (n-1)! \int_0^\infty \frac{e^{-s}}{(1+s)^n} ds.$$

The integral $\int_0^\infty e^{-s}(1+s)^{-n} ds$ is positive, bounded above by $\int_0^\infty (1+s)^{-n} ds = 1/(n-1)$ for $n \geq 2$, and bounded below by

$$\int_0^\infty e^{-s}(1+s)^{-n} ds \geq \frac{1}{e} \int_0^1 (1+s)^{-n} ds = \frac{1}{e(n-1)} \cdot (1 - 2^{1-n}).$$

Hence, $n^2 \kappa_n / n! = n \int_0^\infty e^{-s}(1+s)^{-n} ds \in [c_1, c_2]$ for some absolute constants $0 < c_1 \leq c_2 < \infty$. The bounds in (1.1) then follow by summing in n . $\square$

1.0.2. STFT stability for squares. We now state the main estimate of this subsection, which controls the Fock norm of $F^2 - F(0)^2$ by the $L^2(d\gamma)$ distance from $|F|^2$ to a real constant.

Theorem 1. *There exists an absolute constant $C > 0$ such that for every polynomial $F \in \mathcal{F}^2(\mathbb{C})$, we have*

$$\|F^2 - F(0)^2\|_{\mathcal{F}} \leq C \min_{c \in \mathbb{R}} \| |F|^2 - c \|_{\gamma}. \quad (1.2)$$

The remainder of this subsection is devoted to the proof of this result. We follow the strategy of integrating by parts twice with respect to the weighted measure $(1 + |z|^2)^{-2} d\gamma$, the second time after replacing $|F|^2$ by $|F|^2 - c$.

Step 1: The gradient identity. Recall that, for an entire function F , the Cauchy–Riemann equations imply

$$|\nabla F|^2 = |F'|^2 \quad \text{a.e. on } \mathbb{C}. \quad (1.3)$$

Multiplying by $|F|^2$ one obtains, again from the Cauchy–Riemann equations,

$$|\nabla |F|^2|^2 = 4 |F|^2 |F'|^2 = |(F^2)'|^2 \quad \text{a.e. on } \mathbb{C}. \quad (1.4)$$

Indeed, $(F^2)' = 2FF'$, so $|(F^2)'|^2 = 4|F|^2|F'|^2$, while writing $F = u + iv$ with $u, v \in \mathbb{R}$ and using $u_{x_1} = v_{x_2}$, $u_{x_2} = -v_{x_1}$ gives, on $\{F \neq 0\}$,

$$|\nabla |F|^2|^2 = (2uu_{x_1} + 2vv_{x_1})^2 + (2uu_{x_2} + 2vv_{x_2})^2 = 4(u^2 + v^2)(u_{x_1}^2 + v_{x_1}^2) = 4|F|^2|F'|^2.$$

The zero set of a non-trivial entire function has Lebesgue measure zero, so (1.4) holds a.e. on $\mathbb{C}$. Combining Lemma 1.1 (applied to $H = F^2 - F(0)^2 \in \mathcal{F}_*^2(\mathbb{C})$) with (1.4), we see that

$$\|F^2 - F(0)^2\|_{\mathcal{F}}^2 \leq c_3 \int_{\mathbb{C}} |\nabla |F|^2|^2 \frac{d\gamma(z)}{1 + |z|^2}, \quad (1.5)$$

for some absolute constant $c_3 > 0$.

Step 2: An integral identity. Set $w(z) = (1 + |z|^2)^{-1}/\pi \cdot e^{-|z|^2}$, so that the right-hand side of (1.5) equals $\int_{\mathbb{C}} |\nabla |F|^2|^2 w dm$. Let c be any real number. Two integrations by parts (note that the boundary terms vanish thanks to the Gaussian factor in w and the fact that we are working with polynomials) yield

$$\int_{\mathbb{C}} |\nabla |F|^2|^2 w dm = - \int_{\mathbb{C}} (|F|^2 - c) \Delta |F|^2 w dm + \frac{1}{2} \int_{\mathbb{C}} (|F|^2 - c)^2 \Delta w dm. \quad (1.6)$$

A direct computation shows that there exists an absolute constant C_0 so that

$$|\Delta w(z)| \leq \frac{C_0}{\pi} e^{-|z|^2} \quad \text{for all } z \in \mathbb{C}. \quad (1.7)$$

Indeed, w satisfies the estimate $|\nabla \log w| \leq 2|z| + 2|z|/(1 + |z|^2)$ as well as a similar bound on $\Delta \log w$, so we have $|\Delta w|/w \leq C_0(1 + |z|^2)$, which is simply (1.7). Substituting (1.7) into the second term of (1.6), it follows that

$$\left| \frac{1}{2} \int_{\mathbb{C}} (|F|^2 - c)^2 \Delta w dm \right| \leq \frac{C_0}{2} \| |F|^2 - c \|_{L^2(d\gamma)}^2. \quad (1.8)$$

Step 3: Cauchy–Schwarz on the main term. For the first term in (1.6), Cauchy–Schwarz applied to the splitting $w = (e^{-|z|^2/2}/\sqrt{\pi}) \cdot (e^{-|z|^2/2}/(\sqrt{\pi}(1 + |z|^2)))$ gives

$$\left| \int_{\mathbb{C}} (|F|^2 - c) \Delta |F|^2 w dm \right| \leq \| |F|^2 - c \|_{L^2(d\gamma)} J^{1/2}, \quad (1.9)$$

where

$$J := \int_{\mathbb{C}} (\Delta|F|^2)^2 \frac{d\gamma(z)}{(1+|z|^2)^2}. \quad (1.10)$$

Step 4: Bounding J . We now claim that

$$J \leq C_1 \| |F|^2 - c \|_{L^2(d\gamma)}^2 \quad (1.11)$$

for some absolute constant C_1 . Using the symmetry of the Laplacian and integration by parts twice, we may write

$$J = \int_{\mathbb{C}} \Delta(|F|^2 - c) \Delta|F|^2 \frac{d\gamma}{(1+|z|^2)^2} = \int_{\mathbb{C}} (|F|^2 - c) \Delta(\Delta|F|^2 w_2) dm,$$

where $w_2(z) := (1/\pi)e^{-|z|^2}/(1+|z|^2)^2$. Expanding the outer Laplacian, we obtain

$$\Delta(\Delta|F|^2 w_2) = \Delta^2|F|^2 w_2 + 2\nabla(\Delta|F|^2) \cdot \nabla w_2 + \Delta|F|^2 \Delta w_2.$$

Hence, $J = I_1 + I_2 + I_3$ with the obvious notation. We bound each I_j in turn using the following ingredients.

- (i) the identity $\Delta|F|^2 = 4|F'|^2$ (which gives $\Delta^2|F|^2 = 16|F''|^2$ by another application of (1.4) to F');
- (ii) the pointwise bound $|\nabla(\Delta|F|^2)| = 4|\nabla|F'|^2| \leq 8|F'||F''|$ (which follows from (1.3) applied to F');
- (iii) the bounds $|\nabla w_2| \leq C'_0 (1/\pi)e^{-|z|^2}/(1+|z|^2)^{3/2}$ and $|\Delta w_2| \leq C'_0 (1/\pi)e^{-|z|^2}/(1+|z|^2)$, both of which are derived similarly to (1.7).

With the above in mind, Cauchy–Schwarz against $|F|^2 - c$ then gives, with $d\gamma_k(z) := (1+|z|^2)^{-k} d\gamma(z)$,

$$\begin{aligned} |I_1| &\leq 16 \| |F|^2 - c \|_{L^2(d\gamma)} \left(\int_{\mathbb{C}} |F''|^4 d\gamma_4 \right)^{1/2}, \\ |I_2| &\leq C'_2 \| |F|^2 - c \|_{L^2(d\gamma)} \left(\int_{\mathbb{C}} |F'|^2 |F''|^2 d\gamma_3 \right)^{1/2}, \\ |I_3| &\leq C'_2 \| |F|^2 - c \|_{L^2(d\gamma)} \left(\int_{\mathbb{C}} |F'|^4 d\gamma_2 \right)^{1/2} = C'_2 \| |F|^2 - c \|_{L^2(d\gamma)} J^{1/2}/4, \end{aligned}$$

where in the last identity we used $\Delta|F|^2 = 4|F'|^2$ and the definition of J in (1.10).

The two integrals appearing in $|I_1|$ and $|I_2|$ are controlled by the integral defining J via the following weighted Cauchy integral estimates, whose proofs we postpone to Section 1.0.3.

Lemma 1.2. *There exists an absolute constant $C_3 > 0$ such that for every polynomial $F \in \mathcal{F}^2(\mathbb{C})$,*

$$\begin{aligned} \int_{\mathbb{C}} |F''(z)|^4 d\gamma_4(z) &\leq C_3 \int_{\mathbb{C}} |F'(z)|^4 d\gamma_2(z), \\ \int_{\mathbb{C}} |F'(z)|^2 |F''(z)|^2 d\gamma_3(z) &\leq C_3 \int_{\mathbb{C}} |F'(z)|^4 d\gamma_2(z). \end{aligned}$$

Assuming Lemma 1.2 and recalling that $\int |F'|^4 d\gamma_2 = J/16$, we obtain

$$\max(|I_1|, |I_2|, |I_3|) \leq C_4 \| |F|^2 - c \|_{L^2(d\gamma)} J^{1/2}.$$

Hence, $J \leq 3C_4 \| |F|^2 - c \|_{L^2(d\gamma)} J^{1/2}$, which gives (1.11).

Step 5: Concluding the proof. Substituting (1.8), (1.9) and (1.11) into (1.6) and combining with (1.5), it follows that

$$\|F^2 - F(0)^2\|_{\mathcal{F}}^2 \leq C \| |F|^2 - c \|_{L^2(d\gamma)}^2,$$

which is Theorem 1, since c on the right-hand side is arbitrary. $\square$

1.0.3. The weighted Cauchy integral lemma. We now discuss the proof of the remaining result used in the proof of Theorem 1. We shall employ two main independent ingredients. The first is a pointwise bound for F'' in terms of an average of F' (Lemma 1.3 below) and the second is a comparison of the weights (Lemma 1.4).

Lemma 1.3 (Pointwise Cauchy bound). *Let $F: \mathbb{C} \rightarrow \mathbb{C}$ be entire and let $r > 0$. Then for every $z \in \mathbb{C}$,*

$$|F''(z)|^4 \leq \frac{1}{2\pi r^4} \int_0^{2\pi} |F'(z + r e^{i\theta})|^4 d\theta. \quad (1.12)$$

Proof. The Cauchy integral formula on the circle $\{|w - z| = r\}$ gives

$$F''(z) = \frac{1}{2\pi i} \int_{|w-z|=r} \frac{F'(w)}{(w-z)^2} dw = \frac{1}{2\pi r} \int_0^{2\pi} F'(z + r e^{i\theta}) e^{-i\theta} d\theta.$$

Taking absolute values and using Hölder's inequality, we obtain

$$|F''(z)|^4 \leq \left(\frac{1}{2\pi r} \int_0^{2\pi} |F'(z + r e^{i\theta})| d\theta \right)^4 \leq \frac{1}{(2\pi r)^4} (2\pi)^3 \int_0^{2\pi} |F'(z + r e^{i\theta})|^4 d\theta,$$

which simplifies to (1.12). $\square$

We next record a useful change of variables lemma.

Lemma 1.4. *Let $R \geq 2$. The following statements hold.*

- (i) *For every $z \in \mathbb{C}$ with $|z| \leq R$, every $\theta \in [0, 2\pi)$, and for $w := z + e^{i\theta}$, we have*

$$\frac{e^{-|z|^2}}{(1 + |z|^2)^4} \leq e^{2R+1} (1 + (R+1)^2)^2 \frac{e^{-|w|^2}}{(1 + |w|^2)^2}. \quad (1.13)$$

- (ii) *For every $z \in \mathbb{C}$ with $|z| > R$, every $\theta \in [0, 2\pi)$, and for $w := z + e^{i\theta}/|z|$, the map $z \mapsto w(z, \theta)$ is a C^∞ -diffeomorphism of $\{|z| > R\}$ onto its image, with the absolute value of the Jacobian determinant satisfying*

$$|J(z)| \geq 1 - \frac{1}{R^2}.$$

Moreover, we have

$$\frac{|z|^4 e^{-|z|^2}}{(1 + |z|^2)^4} \leq \frac{e^{2+1/R^2}}{(1 - 2/(1 + R^2))^2} \frac{e^{-|w|^2}}{(1 + |w|^2)^2}. \quad (1.14)$$

Proof. (i) Since $w = z + e^{i\theta}$ we have $|w| \leq |z| + 1 \leq R + 1$ and

$$|z|^2 - |w|^2 = -2 \operatorname{Re}(z e^{-i\theta}) - 1 \leq 2|z| - 1 \leq 2R - 1 < 2R + 1.$$

Hence, $e^{-|z|^2} \leq e^{2R+1} e^{-|w|^2}$. Moreover, $(1 + |z|^2)^{-4} \leq 1$, while $(1 + |w|^2)^2 \leq (1 + (R+1)^2)^2$. Multiplying these three inequalities yields (1.13).

(ii) We begin with the Jacobian estimate. Writing $z = x + iy$ and $w = u + iv$ with $u = x + \cos \theta / |z|$ and $v = y + \sin \theta / |z|$, a direct computation using $\partial_x |z| = x/|z|$ and $\partial_y |z| = y/|z|$ gives

$$\det \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix} = 1 - \frac{\operatorname{Re}(e^{-i\theta} z)}{|z|^3}.$$

Since $|\operatorname{Re}(e^{-i\theta} z)| \leq |z|$, the absolute value of the Jacobian determinant is bounded below by $1 - 1/|z|^2 \geq 1 - 1/R^2$, which is strictly positive for $R \geq 2$. Hence, the map is a C^∞ -local diffeomorphism, and since $|w - z| = 1/|z|$ tends to 0 as $|z| \rightarrow \infty$, it is easy to see that it is injective on $\{|z| > R\}$ provided $R \geq 2$ (which we have fixed).

We now establish the last estimate in the lemma. Since $|w|^2 = |z|^2 + 2 \operatorname{Re}(z e^{-i\theta})/|z| + 1/|z|^2$, we have

$$|z|^2 - |w|^2 = -2 \frac{\operatorname{Re}(z e^{-i\theta})}{|z|} - \frac{1}{|z|^2} \in [-2 - 1/R^2, 2 + 1/R^2],$$

giving $e^{|z|^2 - |w|^2} \leq e^{2+1/R^2}$. For the polynomial weight, we note that $|w| \geq |z| - 1/|z|$, hence on $\{|z| > R\}$ we have

$$\frac{1 + |w|^2}{1 + |z|^2} \geq \frac{1 + (|z| - 1/|z|)^2}{1 + |z|^2} = 1 - \frac{2 - 1/|z|^2}{1 + |z|^2} \geq 1 - \frac{2}{1 + R^2}.$$

Combining the above and using $|z|^4/(1 + |z|^2)^2 \leq 1$, we obtain

$$\begin{aligned} \frac{|z|^4 e^{-|z|^2}}{(1 + |z|^2)^4} &= \frac{|z|^4}{(1 + |z|^2)^2} \cdot \frac{e^{-|z|^2}}{(1 + |z|^2)^2} \leq \frac{e^{|z|^2 - |w|^2} (1 + |w|^2)^2}{(1 + |z|^2)^2} \cdot \frac{e^{-|w|^2}}{(1 + |w|^2)^2} \\ &\leq \frac{e^{2+1/R^2}}{(1 - 2/(1 + R^2))^2} \cdot \frac{e^{-|w|^2}}{(1 + |w|^2)^2}, \end{aligned}$$

which is (1.14). $\square$

Proof of Lemma 1.2. Step 1: Pointwise bounds. Let $r_z := 1$ if $|z| \leq R$ and $r_z := 1/|z|$ if $|z| > R$. Here we may take, for instance, $R = 2$. Lemma 1.3 applied with $r = r_z$ gives, for every $z \in \mathbb{C}$,

$$|F''(z)|^4 d\gamma_4(z) \leq \frac{1}{2\pi r_z^4} \int_0^{2\pi} |F'(z + r_z e^{i\theta})|^4 d\theta \frac{e^{-|z|^2}}{\pi(1 + |z|^2)^4} dm(z).$$

We integrate this inequality first over $\{|z| \leq R\}$, then over $\{|z| > R\}$, freely exchanging the orders of the z - and θ -integration by Fubini's theorem.

Step 2: The inner contribution ($|z| \leq R$). On this region we have $r_z = 1$, so

$$\frac{1}{r_z^4} \frac{e^{-|z|^2}}{(1 + |z|^2)^4} \leq e^{2R+1} (1 + (R+1)^2)^2 \frac{e^{-|w|^2}}{(1 + |w|^2)^2}$$

by Lemma 1.4 (i) with $w = z + e^{i\theta}$. The map $z \mapsto z + e^{i\theta}$ is a translation, hence has unit Jacobian, and sends $\{|z| \leq R\}$ into $\{|w| \leq R+1\}$. Therefore,

$$\int_{|z| \leq R} \frac{dm(z)}{2\pi^2 r_z^4} \int_0^{2\pi} |F'(z + e^{i\theta})|^4 \frac{e^{-|z|^2}}{(1 + |z|^2)^4} d\theta \leq e^{2R+1} (1 + (R+1)^2)^2 \int_{\mathbb{C}} |F'(w)|^4 d\gamma_2(w),$$

where the right-hand side has been enlarged from $\{|w| \leq R+1\}$ to all of $\mathbb{C}$ since the integrand is nonnegative.

Step 3: The outer contribution ($|z| > R$). On this region we have $r_z = 1/|z|$, so

$$\frac{1}{r_z^4} \frac{e^{-|z|^2}}{(1+|z|^2)^4} = \frac{|z|^4 e^{-|z|^2}}{(1+|z|^2)^4}.$$

By applying Lemma 1.4 (ii) to this expression with $w = z + e^{i\theta}/|z|$, we see that

$$\frac{e^{2+1/R^2}}{(1-2/(1+R^2))^2} \frac{e^{-|w|^2}}{(1+|w|^2)^2},$$

and the map $z \mapsto w$ is a diffeomorphism with the absolute value of its Jacobian determinant $\geq 1 - 1/R^2$. Hence, the change of variables produces an extra factor which is at most $(1 - 1/R^2)^{-1}$.

We therefore have

$$\begin{aligned} \int_{|z|>R} \frac{dm(z)}{2\pi^2 r_z^4} \int_0^{2\pi} |F'(z + e^{i\theta}/|z|)|^4 \frac{e^{-|z|^2}}{(1+|z|^2)^4} d\theta \\ \leq \frac{e^{2+1/R^2}}{(1-1/R^2)(1-2/(1+R^2))^2} \int_{\mathbb{C}} |F'(w)|^4 d\gamma_2(w). \end{aligned}$$

Note that the factor $|z|^4$ produced by r_z^{-4} has already been absorbed into (1.14) via the elementary bound $|z|^4/(1+|z|^2)^2 \leq 1$.

Step 4: Concluding the proof. Combining Steps 2 and 3, we have

$$\int_{\mathbb{C}} |F''(z)|^4 d\gamma_4(z) \leq A_1 \int_{\mathbb{C}} |F'(w)|^4 d\gamma_2(w),$$

with A_1 an absolute constant. This is the first inequality in Lemma 1.2. We finally write the integrand of $\int_{\mathbb{C}} |F'|^2 |F''|^2 d\gamma_3$ as

$$(|F'(z)|^2 (1+|z|^2)^{-1}) (|F''(z)|^2 (1+|z|^2)^{-2}) \frac{e^{-|z|^2}}{\pi} dm(z)$$

and apply Cauchy-Schwarz to obtain

$$\int_{\mathbb{C}} |F'|^2 |F''|^2 d\gamma_3 \leq \left(\int_{\mathbb{C}} |F'|^4 d\gamma_2 \right)^{1/2} \left(\int_{\mathbb{C}} |F''|^4 d\gamma_4 \right)^{1/2}.$$

Substituting the first bound into the second factor, we obtain that

$$\int_{\mathbb{C}} |F'|^2 |F''|^2 d\gamma_3 \leq \left(\int_{\mathbb{C}} |F'|^4 d\gamma_2 \right)^{1/2} \sqrt{A_1} \left(\int_{\mathbb{C}} |F'|^4 d\gamma_2 \right)^{1/2} = \sqrt{A_1} \int_{\mathbb{C}} |F'|^4 d\gamma_2,$$

which concludes the proof. $\square$